MILITARY SPECIFICATION

LIGHT, BED, HOSPITAL

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers an adjustable bed light for use in r dical treatment facilities.

2. APPLICABLE DOCUMENTS

2.1 Issues of documents. The following documents of the issue in effect on date of invitation for bids or request for proposals, form a part of this specification to the extent specified herein:

SPECIFICATIONS

FEDERAL

TT-C-490	-	Cleaning Method and Pretreatment of Ferrous Surfaces for Organic Coatings.
PPP-B-585	-	Boxes, Wood, Wirebound.
PPP-B-601	-	Boxes, Wood, Cleated-Plywood.
PPP-B-621	-	Boxes, Wood, Nailed and Lock-Corner.
PPP-B-636	-	Boxes, Shipping, Fiberboard.

MILITARY

MIL-P-116	-	Preservation-packaging, Methods of.
MIL-C-5541	-	Chemical Conversion Coatings on Aluminum and Aluminum Alloys.
MIL-A-8625	-	Anodic Coatings for Aluminum and Aluminum Alloys.
MIL-L-10547	-	Liners, Case, and Sheet, Overwrap; Water-vaporproof, or Waterproof, Flexible.

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Headquarters, Defense Personnel Support Center, ATTN: Directorate of Medical Materiel, DPSC-ATT, 2800 South 20th Street, Philadelphia, PA 19101, by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

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for proposal the issue in effect on date of invitation for bids or request

STANDARDS

FEDERAL

FED-STD-66 - Steel, Chemical Composition and Hardenability.

MILITARY

MIL-STD-105 - Sampling Procedures and Tables for Inspection by Attributes.
MIL-STD-129 - Marking for Shipment and Storage.

(Copies of specifications and standards required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

2.2 Other publications. The following documents form a part of this specification to the extent specified herein. Unless otherwise specified, the issue in effect on date of invitation for bids or request for proposals shall apply.

AMERICAN NATIONAL STANDARDS INSTITUTE (ANSI)

C73.11 - Dimensions of Caps, Plugs and Receptacles 125 Volts, 15 Amperes, 2 Pole, 3 Wire, Grounding Type.

(Copies of ANSI Standards may be obtained upon application to the American National Standards Institute Inc., 1430 Broadway, New York, NY 10018.)

UNDERWRITERS' LABORATORIES, INC. (UL)

UL 153 - Portable Electric Lamps.
UL 544 - Medical and Dental Equipment.

(Applications for copies should be addressed to the Underwriter's Laboratories, Inc., 207 East Ohio Street, Chicago, IL 60611, 333 Pfingsten Rd., Northbrook, IL 60062, 1285 Walt Whitman Rd., Melville, L.I., NY 11746, 1655 Scott Blvd., Santa Clara, CA 95050.)

3. REQUIREMENTS

3.1 First article. When specified in the procurement document, the contractor shall furnish sample unit(s) for first article inspection and approval (see 4.1.2.1 and 6.2).

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3.2 Fire and casualty hazards.

3.2.1 The contractor shall submit proof, prior to delivery, that the items to be delivered under the contract conform to the applicable requirements of Underwriters' Laboratories (UL) Inc., Standard of Safety, UL 153 and the applicable portions of UL 544. Proof shall consist of one of the following:

A. UL label affixed to the product, provided the contractor agrees to have Underwriters' Laboratories, Inc. forward a copy of the UL report directly to the Contracting Officer, upon the latter's request.

B. Laboratory report from a nationally recognized independent laboratory. If an independent laboratory is used, it must be adequately equipped and competent to determine compliance with UL standards. The Government reserves the right to require a facilities report and to conduct a facilities survey. Any laboratory report submitted shall include actual test methods, data taken, and results of tests. Detailed findings of examinations necessary to determine compliance shall also be included. The test report shall also have a section that describes the test sample in such a way that the report can be used to insure identity of production items with the test information such as materials used, spacings between electrical components, types of bushings, photographs, wiring diagrams, sketches, and parts list, all appropriately identified. Compliance with this paragraph does not excuse compliance with all other requirements of the contract and specifications.

3.2.2 Compliance with the above requirements regarding fire and casualty hazards does not absolve the contractor from complete compliance with the requirements of this specification in order to secure the acceptance of his commodity.

3.3 Materials. All material used in the construction of the light shall be of high quality throughout, suitable in every respect for the purpose intended.

3.3.1 Steel. Standard commercial tolerances shall apply to thicknesses of steel specified herein.

3.3.1.1 Steel sheet. Steel sheet used in the construction of the light shall be cold rolled, AISI 1000 series of FED-STD-66, 1010 or higher.

3.3.1.2 Steel tubing. Steel tubing used in the construction of the light shall be cold rolled, AISI 1000 series of FED-STD-66, 1020 or higher.

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3.3.2 Aluminum. Aluminum used in the construction of the light shall be aluminum alloy.

3.4 Design and construction. The light shall be suitable for use as a reading light and as a patient-examining light (see figure 1). Sufficient slack shall be provided in the internal wiring so that no tension shall be developed as the assembly is articulated through any allowable adjustment. Wiring shall also be protected against rubbing or fraying, especially in the joints. The light shall be designed to operate from a 115 volt power source.

3.4.1 Base. The base housing shall be constructed of steel, at least .020 inch thick (No. 25 Manufacturers' Standard gage). It shall be capable of being opened with a common screwdriver to permit maintenance. An on-off switch shall be mounted on the control housing or on the reflector. The mounting stem shall be nickel-plated steel rod, 1/2 ± 1/16 inch in diameter, tapered at the bottom. The stem may be supplied with a flange at the top. The stem shall be designed to extend at least 1-5/8 inches into the intravenous socket of the bed or mounting bracket. The assembly shall be constructed so that the extension arms of the light can rotate at least 60 degrees to either side of center in the horizontal plane. If the point of rotation is above the base housing, the mounting stem shall be equipped with a steel piece which extends over the top and down the back of the headboard, parallel to the stem. This steel piece shall have a knurled setscrew, which passes through the piece to provide pressure on the back of the headboard, to prevent the base from turning in the intravenous socket or mounting bracket (see figure 1). Means shall be provided to prevent the tip of the thumbscrew from marring the finish of the bed. Alternatively, the rotation may be accomplished by means of the mounting stem itself. In this case, the stem shall fit into the intravenous socket with sufficient friction to remain in position after adjustment, but shall be capable of being rotated horizontally without requiring undue force.

3.4.2 Extension arms. The extension arms shall consist of two rigid arms, made from steel tubing at least .035 inch thick (No. 20 Birmingham Wire gage), joined by a vertically adjustable elbow joint. Either or both arms may be of single or double-rod construction. The elbow joint shall maintain any setting from fully retracted (with reflector touching the inner arm) to fully extended. The joint which connects the inner arm to the control unit shall allow the inner arm to retain any position from horizontal through vertical. When fully extended, the outside of the reflector shall be at least 30 inches from the base axis. The arms shall remain rigid in all positions, without sagging or other distortions. Any oscillation or vibration shall cease in less than 10 seconds after adjustment. After being tested as specified in 4.4.1, the light shall retain any position to which it is adjusted, with a maximum deviation of one inch in any direction, and shall exhibit no distortion. The joints shall retain their settings by means of balance mechanisms or compression-spring loaded friction joints; thumbscrews, wing nuts, or other methods of locking the adjustments shall not be employed.

3.4.3. Reflector assembly. The reflector assembly shall consist of the reflector, the reading lamp socket, and the socket support bracket (if required). It may also include the switch.

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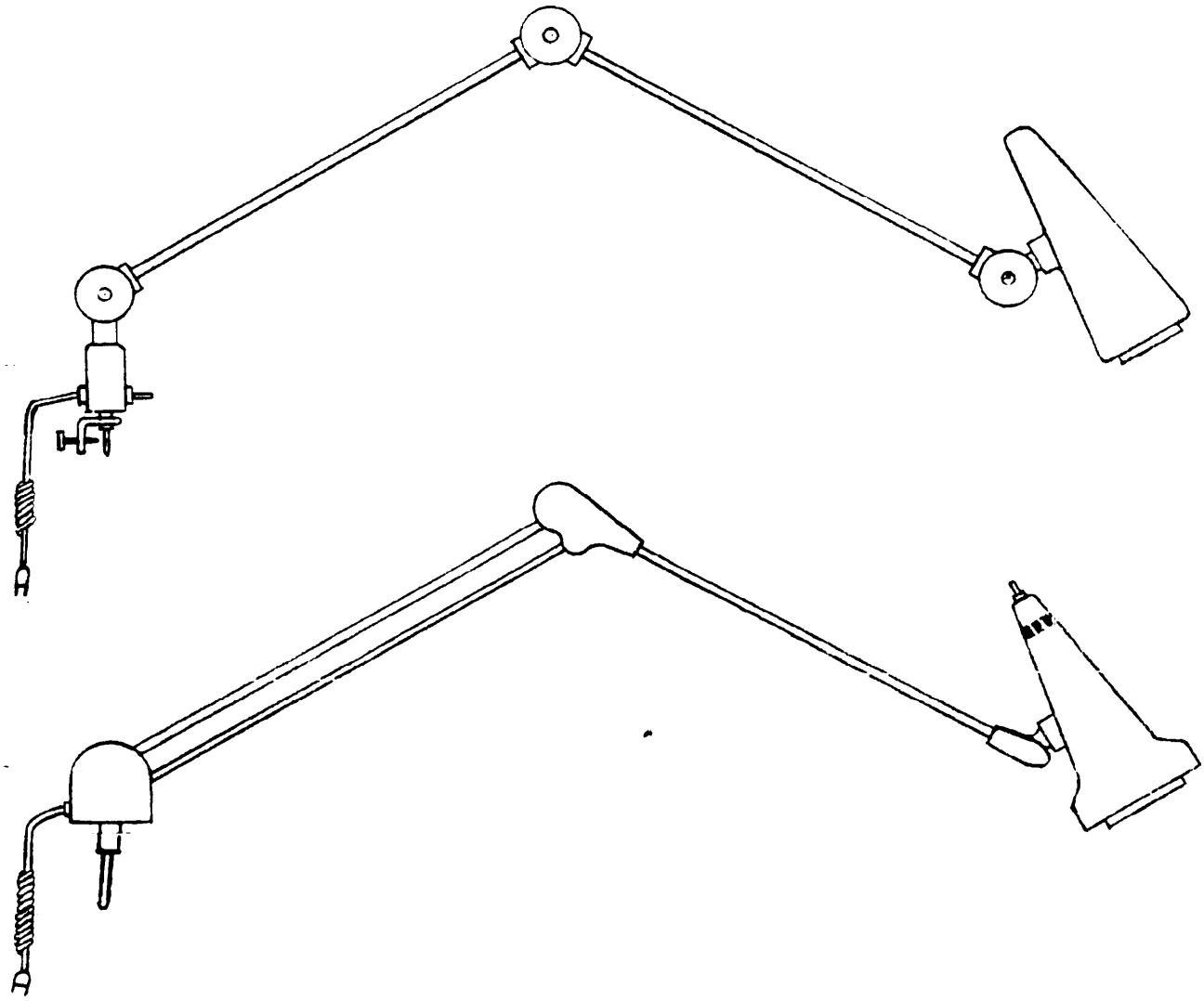


FIGURE I. Light, bed. Typical configurations.

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contract or purchase order. The supplier is responsible for the performance

3.4.3.1 Reflector. The reflector shall be made of aluminum at least .030 inch thick. It shall be basically cone shaped, not more than 8 inches in length and not more than 7 inches in diameter. The reflector shall be of double shell construction. The interior of the reflector shall be anodized. The lamp shall be recessed within the reflector. By means of venting, or a combination of venting and a guard, across the face of the reflector, the temperature of the outside of the reflector shall be held below 50° C (122° F) at any point when tested as specified in 4.4.4. The light shall not cause the temperature of the thermocouple to exceed 149° C (300° F) when tested as specified in 4.4.5. Movement of the reflector about the outer end of the outer extension arm shall have the freedom of motion shown in figures 2a, 2b, and 2c. This shall be achieved by any joint, or combination of joints, which meets all applicable requirements of this specification.

3.4.3.2 Socket. The socket shall be porcelain, medium base type, rated for at least 250 watts, 250 volts. It shall conform to the requirements of UL 153. It shall be fastened to the reflector, either directly or by means of a bracket. The socket shall be capable of being detached from the reflector by removing not more than four screws. The socket shall be mounted in the reflector so that it cannot be rotated when removing or inserting a light bulb.

3.4.3.3 Incandescent lamp. The light shall accommodate a medium screw base incandescent lamp, type A-19 bulb, 60 watts.

3.4.4 Switch. A push-canopy or toggle switch, conforming to UL 153, shall be mounted on the control housing or on the reflector. The ungrounded conductor shall be connected to the switch, so that the screw shell and the center contact of the socket shall be electrically dead when the switch is in the "off" position.

3.4.5 Cord and plug. Cord shall be flexible, 3-conductor, type SJO, SJT, SJTO, or at least as serviceable, not less than No. 18 American Wire gage, 9 feet in length measured from the plug to the base, and equipped with a Hospital Grade, grounded, three-prong plug as specified in UL 544, which is compatible with the receptacle specified in American National Standards Institute (ANSI) Standard C73.11. The grounding wire connector shall be secured to the frame or switch mechanism enclosure by a screw that need not be removed during any servicing operation not involving the power supply cord, or by other equivalent means. Ordinary solder alone shall not be used for securing the grounding conductor. Suitable strain relief shall be provided in accordance with UL 544.

3.5 Dielectric strength. No electrical breakdown shall occur when the light is tested as specified in 4.4.2.

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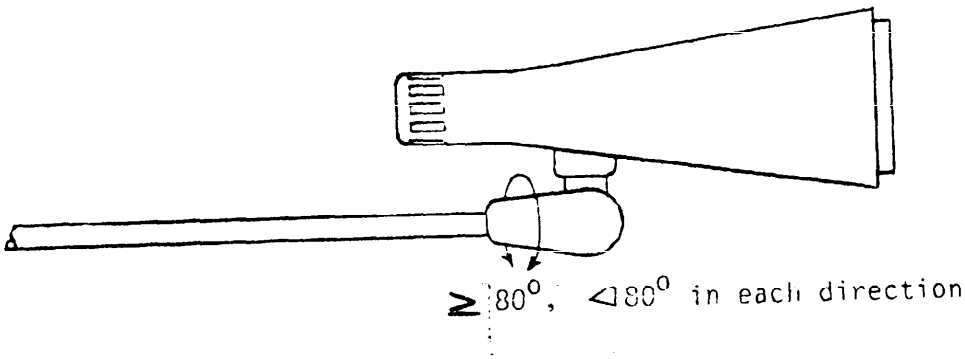


FIGURE 2a - Movement of Reflector

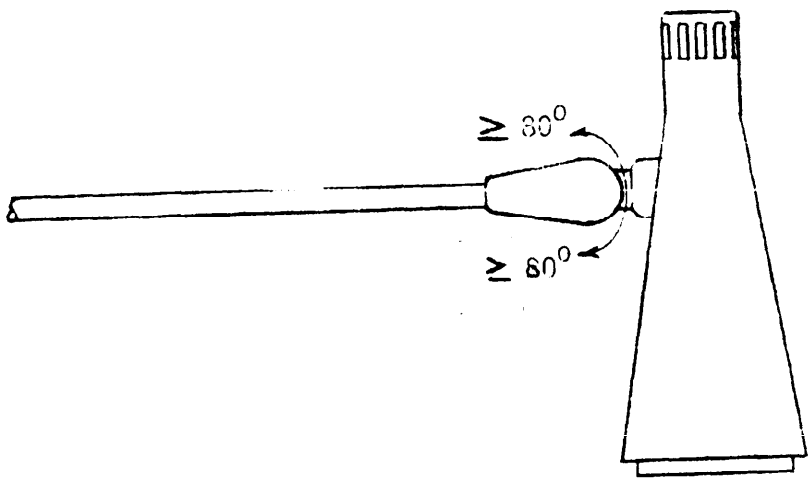


FIGURE 2b - Movement of Reflector

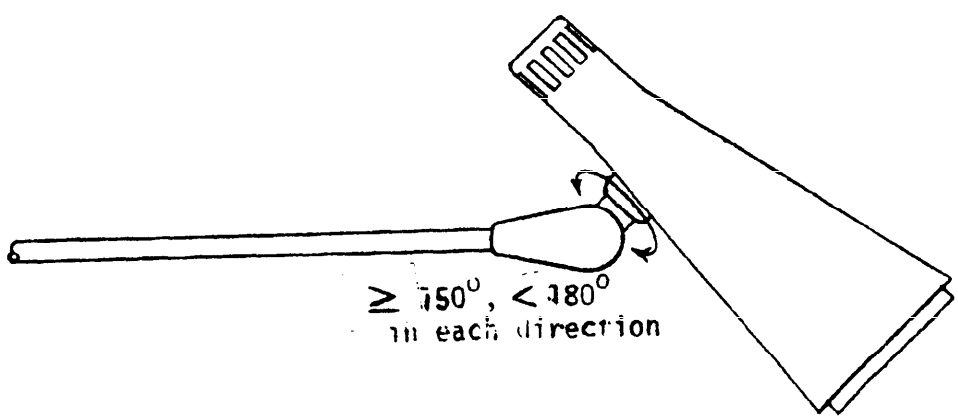


FIGURE 2c - Movement of Reflector

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3.6 Finish.

3.6.1 Cleaning and priming.

3.6.1.1 Steel surfaces. Exterior steel surfaces shall be cleaned in accordance with a method prescribed in TT-C-490. Coating shall conform to type II of TT-C-490.

3.6.1.2 Aluminum surfaces. Exterior aluminum surfaces shall be prepared in accordance with a method prescribed in MIL-A-8625 or MIL-C-5541 before being enameled.

3.6.2 Enamel. After cleaning, the exterior steel and aluminum surfaces shall be painted with a metallic gloss baking enamel; color shall be natural or statuary bronze. Enamel shall be baked in accordance with the manufacturers' directions. Finished surface shall be a uniform, continuous film, free from dried overspray, runs, sags, blisters, orange peel, and other imperfections.

3.6.3 Plated surfaces. Plated surfaces shall have a smooth, even, bright finish.

3.6.4 Plastic surfaces. Exposed plastic surfaces shall match or harmonize with the color of the enameled surfaces.

3.7 Identification marking. Each lamp shall be permanently and legibly marked with the contract or purchase order number, and the name or registered trademark of the manufacturer.

3.8 Service data. The contractor shall furnish, with each unit, two copies each of a technical manual which contains complete instructions for installation, operation and maintenance, and lists of component parts.

3.9 Workmanship. Workmanship shall be first class throughout. Lights shall be free from defects which detract from their appearance or impair their serviceability.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

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4.1.1 Records. Records of examinations and tests performed by or for the contractor shall be maintained by the contractor and made available to the Government, upon the Government's request, at any time, or from time to time, during the performance of the contract and for a period of three years after delivery of the supplies to which such records relate.

4.1.2 Inspection. Inspection, as used in this specification, is defined as both examination (such as visual or auditory investigation without the use of special laboratory appliances or procedures) and testing (determination by technical means of physical and chemical properties) of the item.

4.1.2.1 Classification of inspection. The inspection requirements specified herein are classified as follows:

a. First article inspection.
b. Quality conformance inspection.

4.1.2.1.1 First article inspection. First article inspection shall be performed on the bed light when a first article sample is required (see 3.1). This inspection shall include examination of 4.3 and tests of 4.4.

4.1.2.1.2 Quality conformance inspection. Quality conformance inspection shall include the examination of 4.3 and tests of 4.4.

4.1.3 Certificates of quality. Certificates of quality, supplied by the manufacturer of the raw materials may be furnished in lieu of actual performance of such testing by the contractor, provided lot identity has been maintained and can be demonstrated to the Government. The certificate shall include the name of the contractor, the contract number, the name of the manufacturer or supplier, the NSN, the Item Identification, the name of the component or material, the lot number, the lot size, the sample size, the date of testing, the test method, individual test results, and the specification requirements.

4.2 Sampling. The unit of product for sampling purposes shall be one completely assembled light.

4.2.1 For examination. Sampling for examination shall be conducted in accordance with MIL-STD-105 and as indicated in table I.

TABLE I. Sampling for examination.

	Inspection level	AQL (percent defective)
For visual examination		
Major defects	II	1.0
Minor defects	II	4.0
For dimensional examination	S-1	2.5

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4.2.2 For tests. Sampling for tests shall be conducted in accordance with MIL-STD-105, and as indicated in table II. Acceptance level shall be zero for all sample sizes.

TABLE II. Sampling criteria for tests.

Component (unit of product)	Characteristic	Requirement	Test procedure	Inspection level
Light	Stability	3.4.2	4.4.1	S-2
	Dielectric strength	3.5	4.4.2	100%
	Polarity	3.4.4	4.4.3	100%
Reflector	Temperature	3.4.3.1	4.4.4	1/
	Fire hazard	3.4.3.1	4.4.5	1/

1/ One unit per contract.

4.3 Examination. The light shall be examined to determine compliance with all requirements contained in this specification.

4.3.1 Classification of defects. Examination shall be conducted in accordance with table III. Examination shall not be limited to the classified possible defects listed below.

TABLE III. Classification of defects.

Categories	Defects
Major:	
101	Component part missing.
102	Any adjustable part fails to move easily.
103	Component parts in assembled light cracked, torn or ruptured.
104	Enameled surfaces have unenameled or corroded areas.
105	Ground wire not adequately secured to frame or enclosure.
106	Interior of wire raceway, including joints, has burrs or sharp edges.
107	Wire strain relief inadequate in any position of adjustment.
108	Wire is strained at any joint, in the reflector, or at any point of articulation.
109	Ungrounded conductor has insulation colored white, natural gray, or green.
Minor:	
201	Assembled light exhibits dirt, oil, grease, chips, or foreign matter.
202	Identification marking not as specified.

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4.3.2 Dimensional examination. The light shall be examined for defects in dimensions. Any dimension not within the tolerance specified herein shall be classified as a defect.

4.3.3 Packaging inspection. The inspection of the packaging and interior package marking shall be in accordance with the group A and B quality conformance inspection requirements of MIL-P-116. The inspection of the packing and marking for shipment and storage shall be in accordance with the quality assurance provisions of the applicable container specification and the marking requirements of MIL-STD-129.

4.4 Tests. Tests shall be conducted to determine compliance with specification requirements.

4.4.1 Stability test. The light shall be fully retracted, with the reflector, or the reflector joint, in contact with the inner extension arm and the reflector pointing straight down. The arms shall then be fully extended. The reflector shall be rotated to the full up position, and also tilted (at the reflector joint) to the full up position. The light shall then be returned to its original configuration. This test shall be repeated for 50 complete cycles. The lamp shall then be examined for distortion and for failure to retain any setting. The internal wiring shall be examined for evidence of strain or damage to the insulation.

4.4.2 Dielectric strength. The light shall be tested for dielectric withstand as specified in UL 544.

4.4.3 Polarity. A continuity check shall be made to determine if the switch is wired directly to the ungrounded conductor terminal on the plug, as specified in UL 153.

4.4.4 Temperature test. The light, equipped with a 60 watt bulb, shall be left on for at least three hours with the illumination directed straight down. After three hours, the temperature of the outside of the reflector shall be measured with a thermocouple or an optical pyrometer with an accuracy of $\pm$ 5 percent.

4.4.5 Fire hazard test. The light, equipped with a 60 watt bulb, shall be adjusted so that the illumination is straight down. The light shall be turned on and the front of the reflector pushed, with at least 10 pounds of force, into direct contact with a cotton bedsheet placed on a firm mattress or pillow. (The pressure need not be maintained after initial application.) A thermocouple shall be placed under the sheet within the circumference of the reflector. The light shall be left on for one hour, and the temperature shall be recorded at the end of the hour.

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5. PACKAGING

5.1 Packaging. Packaging shall be level A or C as specified (see 6.2).

5.1.1 Level A.

5.1.1.1 Unit packaging. Each light shall be packaged in a box of appropriate size and design constructed in accordance with PPP-B-636, type CF, class domestic. Box shall be designed with interior fittings or shall contain sufficient cushioning material to insure aginst shifting of or damage to the contents. Box shall be closed as specified in the appendix of the box specification. Packaging shall conform to MIL-P-116, method III.

5.1.2 Level C. Lights shall be packaged in standard commercial containers of the size and kind commonly used, which will afford the degree of protection required for shipment and use of the product for its intended purpose.

5.2 Packing. Packing shall be level A, B or C as specified (see 6.2).

5.2.1 Level A. Four (4) lights shall be packed in an exterior container designed for a type 2 load and constructed in accordance with PPP-B-585, class 3, style 3; PPP-B-601, overseas type; PPP-B-621, class 2; or PPP-B-636, class weather-resistant. Closure and strapping shall be as specified in the appendix of the applicable box specification.

5.2.1.1 Case liner. Each level A wood box shall be lined with a waterproof case liner conforming to MIL-L-10547. Closure and sealing shall conform to applicable paragraphs of appendix thereto. Case liner shall not be required for fiberboard boxes. Each fiberboard box shall be waterproofed as specified in the appendix of PPP-B-636.

5.2.2 Level B. Four (4) lights shall be packaged in an exterior container designed for a type 2 load and constructed in accordance with PPP-B-585, class 1, style 3; PPP-B-601, domestic type; PPP-B-621, class 1; of the applicable box specification. Closure of fiberboard boxes shall conform to method II as specified in PPP-B-636.

5.2.3 Level C. The subject commodity shall be packed in substantial commercial containers of the type, size, and kind commonly used for the purpose, so constructed as to insure acceptance and safe delivery by common or other carriers, at the lowest rate, to point of delivery called for in the contract or purchase order.

5.3 Marking. Each unit package and exterior container shall be marked as specified in MIL-STD-129.

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6. NOTES

6.1 Intended use. The bed light is intended to be mounted on the headboard of a hospital bed for use as a reading or examining light.

6.2 Ordering data. Procurement documents should specify the following:

a. Title, number and date of this specification.
b. Selection of applicable levels of packaging and packing (see 5.1 and 5.2).
c. When a first article is required (see 3.1, 4.1.2.1 and 6.3).

6.3 First article. When a first article is required, it shall be tested and approved under the appropriate provisions of the Armed Services Procurement Regulations. The contract or purchase order specifies the quantity, procedure for delivery and other specific instructions for inspection and approval of the first article.

6.4 This specification covers the following item appearing in the Federal Supply Catalog:

National Stock Number	Item Identification
6230-00-145-1407	LIGHT, BED, 115 volt, single phase, AC

Custodians:
Army - MD
Navy - MS
Air Force - 03

Review activity:
DOD - MB

Preparing activity:
DLA-DM

Project No. 6230-0229

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

OMB Approval
No. 22-R255

INSTRUCTIONS: The purpose of this form is to solicit beneficial comments which will help achieve procurement of suitable products at reasonable cost and minimum delay, or will otherwise enhance use of the document. DoD contractors, government activities, or manufacturers/ vendors who are prospective suppliers of the product are invited to submit comments to the government. Fold on lines on reverse side, staple in corner, and send to preparing activity. Comments submitted on this form do not constitute or imply authorization to waive any portion of the referenced document(s) or to amend contractual requirements. Attach any pertinent data which may be of use in improving this document. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity.

DOCUMENT IDENTIFIER AND TITLE
MIL-L-37502 LIGHT, BED, HOSPITAL

NAME OF ORGANIZATION AND ADDRESS

CONTRACT NUMBER

MATERIAL PROCURED UNDER A
☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE DOCUMENT CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?
   A. GIVE PARAGRAPH NUMBER AND WORDING.

   B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

2. COMMENTS ON ANY DOCUMENT REQUIREMENT CONSIDERED TOO RIGID

3. IS THE DOCUMENT RESTRICTIVE?
   ☐ YES ☐ NO *(If "Yes", in what way?)*

4 REMARKS

SUBMITTED BY *(Printed or typed name and address - Optional)*

TELEPHONE NO.

DATE

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